



华中科技大学

Huazhong University of Science and Technology



华中科技大学

网络空间安全学院

School of Cyber Science and Engineering, HUST

实验三 SLAM地图构建和导航

华中科技大学 2021 ROS创新实践教学



日期：2021.03



内容：SLAM地图与导航功能

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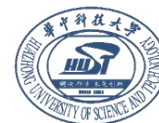
概述

了解 SLAM

连接 小车

建立 SLAM地图

使用 导航功能

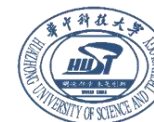


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SLAM (simultaneous localization and mapping),也称为CML (Concurrent Mapping and Localization), **即时定位与地图构建**, 或并发建图与定位。

问题可以描述为：将一个机器人放入未知环境中的未知位置，是否有办法让机器人一边逐步描绘出此环境完全的地图。所谓完全的地图 (a consistent map) 是指不受障碍行进到房间可进入的每个角落。

SLAM最早由Smith、Self和Cheeseman于1988年提出。由于其重要的理论与应用价值，被很多学者认为是实现**真正全自主移动机器人**的关键。



参考实验二小车连接方法

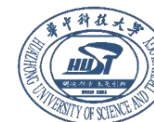
【主机】

\$ roscore # 启动roscore并检验网络配置是否成功

【Burger】

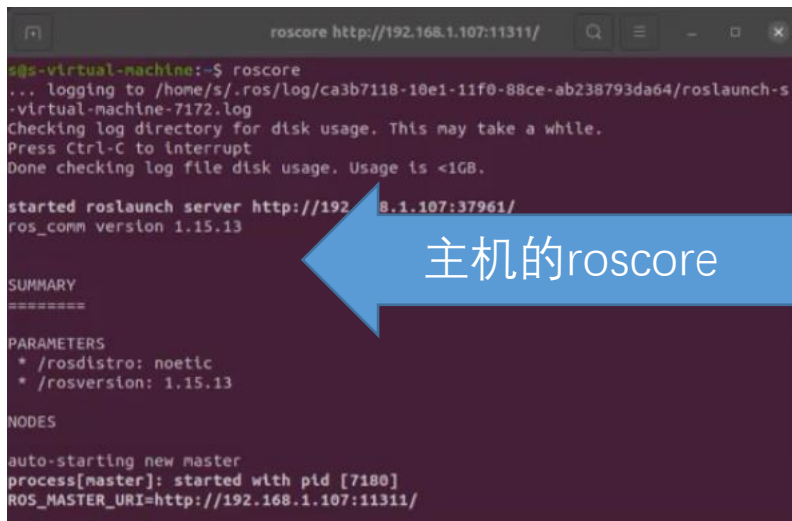
\$ ssh -Y pi@Burger的IP # ssh连接从机

\$ roscore # 启动roscore并检验网络配置是否成功



正确结果：

roscore不报Error。



```
roscore http://192.168.1.107:11311/

s@s-virtual-machine:~$ roscore
... logging to /home/s/.ros/log/ca3b7118-10e1-11f0-88ce-ab238793da64/roslaunch-s-virtual-machine-7172.log
Checking log directory for disk usage. This may take a while.
Press Ctrl-C to interrupt
Done checking log file disk usage. Usage is <1GB.

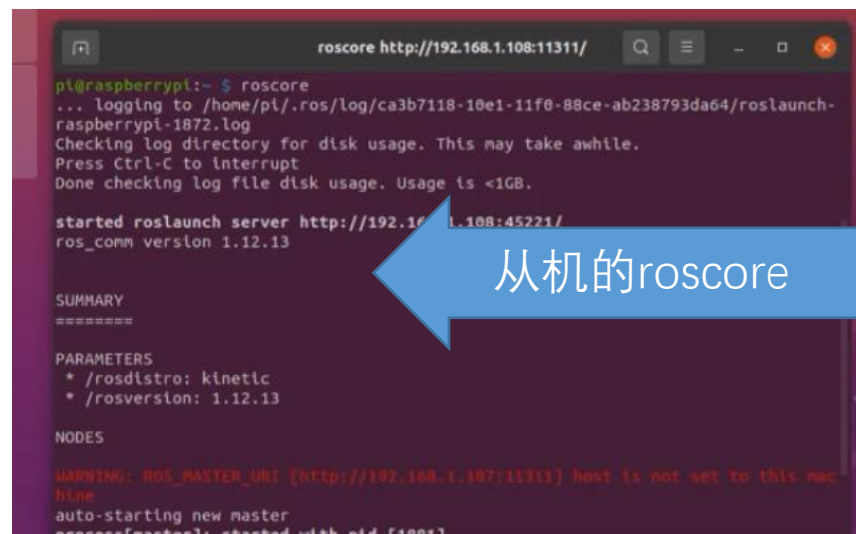
started roslaunch server http://192.168.1.107:37961/
ros_comm version 1.15.13

SUMMARY
=====
PARAMETERS
 * /rostdistro: noetic
 * /rosversion: 1.15.13

NODES

auto-starting new master
process[master]: started with pid [7180]
ROS_MASTER_URI=http://192.168.1.107:11311/
```

主机的roscore



```
roscore http://192.168.1.108:11311/

pi@raspberrypi:~$ roscore
... logging to /home/pi/.ros/log/ca3b7118-10e1-11f0-88ce-ab238793da64/roslaunch-raspberrypi-1872.log
Checking log directory for disk usage. This may take awhile.
Press Ctrl-C to interrupt
Done checking log file disk usage. Usage is <1GB.

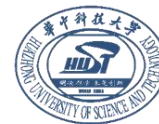
started roslaunch server http://192.168.1.108:45221/
ros_comm version 1.12.13

SUMMARY
=====
PARAMETERS
 * /rostdistro: kinetic
 * /rosversion: 1.12.13

NODES

WARNING: ROS_MASTER_URI [http://192.168.1.107:11311] host is not set to this machine
auto-starting new master
process[master]: started with pid [1881]
```

从机的roscore



本关涉及的指令：

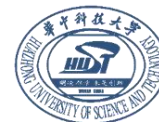
执行slam启动命令

roslaunch turtlebot3_slam turtlebot3_slam.launch

启动键盘控制teleop命令

roslaunch turtlebot3_teleop turtlebot3_teleop_key.launch

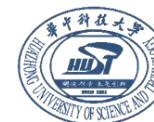
保存SLAM地图至~/map : rosrun map_server map_saver -f ~/map



小车端运行命令

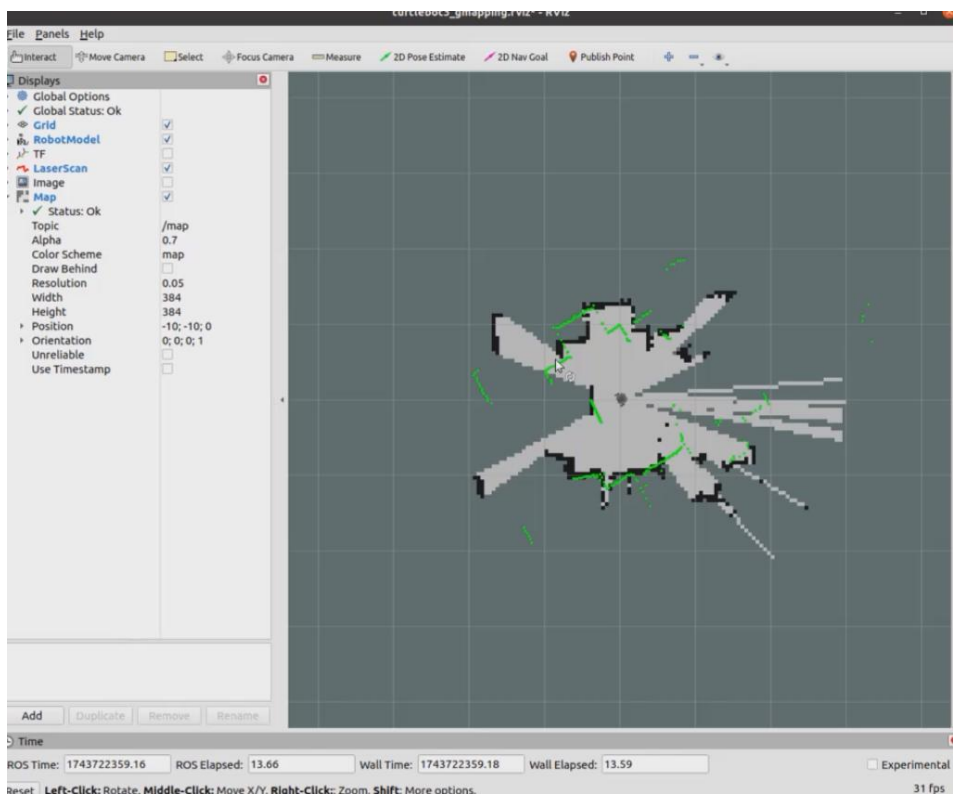
turtlebot3_bringup turtlebot3_robot.launch

```
pi@raspberrypi:~ $ roslaunch turtlebot3_bringup turtlebot3_robot.launch
... logging to /home/pi/.ros/log/ca3b7118-10e1-11f0-88ce-ab238793da64/roslaunch-
raspberrypi-1987.log
Checking log directory for disk usage. This may take awhile.
Press Ctrl-C to interrupt
Done checking log file disk usage. Usage is <1GB.
```



主机端运行slam程序

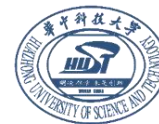
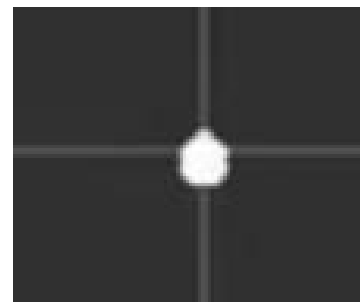
```
roslaunch turtlebot3_slam turtlebot3_slam.launch
```



正确结果
出现地形及小车

错误结果
只有一个白色的车
/没有地图
/命令中止

检查之前的指令结果
+ 解决该指令的
Warnig和Error



主机端启动键盘控制teleop命令

roslaunch turtlebot3_teleop turtlebot3_teleop_key.launch

```
qgs-virtual-machine:~$ roslaunch turtlebot3_teleop turtlebot3_teleop_key.launch
... logging to /home/s/.ros/log/ca3b7118-10e1-11f0-88ce-ab238793da64/roslaunch-s
-virtual-machine-7461.log
Checking log directory for disk usage. This may take a while.
Press Ctrl-C to interrupt
Done checking log file disk usage. Usage is <1GB.

started roslaunch server http://192.168.1.107:38697/

SUMMARY
=====
PARAMETERS
 * /model: burger
 * /roscdistro: noetic
 * /rosversion: 1.15.13
NODES
 /
  turtlebot3_teleop_keyboard (turtlebot3_teleop/turtlebot3_teleop_key)
ROS_MASTER_URI=http://192.168.1.107:11311
```

正常结果
无报错

- # 操作：操控机器人把实验场地都浏览一遍
- # 1、扫描过程中需要注意尽量不要使机器人与环境产生接触。
- # 2、不要人为在机器人附近活动。
- # 3、不要将机器人速度提升到太快，太快的移动速度会导致环境探测信号量减少。
- # 4、按照规范操作，以便得到正确的地形数据。

保存SLAM地图至~/map。将生成map.pgm/map.yaml两个文件。

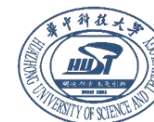
roslaunch map_server map_saver -f ~/map

如果报错命令找不到，或者命令出错，试试：

sudo apt-get install ros-noetic-map-server

```
s@ubuntu-virtual-machine:~$ roslaunch map_server map_saver -f ~/map
[ INFO] [1743722455.422769651]: Waiting for the map
[ INFO] [1743722455.736773935]: Received a 384 X 384 map @ 0.050 m/pix
[ INFO] [1743722455.737313667]: Writing map occupancy data to /home/s/map.pgm
[ INFO] [1743722455.758862964]: Writing map occupancy data to /home/s/map.yaml
[ INFO] [1743722455.759718487]: Done
```

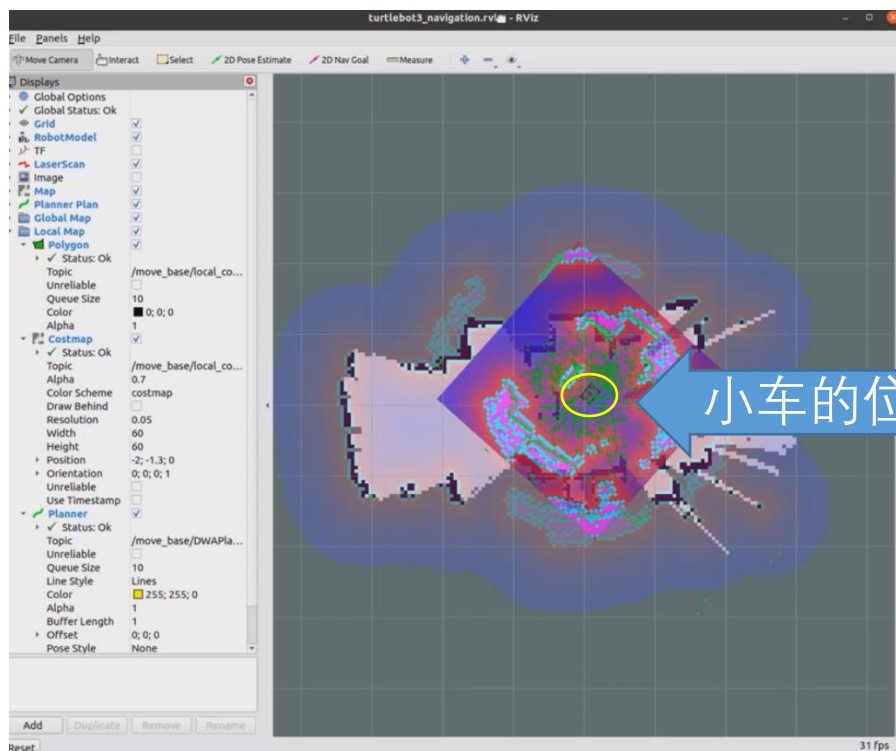
正常结果
无报错



读取测量好的地图数据~/map.yaml

`roslaunch turtlebot3_navigation turtlebot3_navigation.launch`

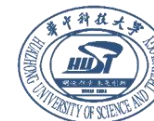
`map_file:=$HOME/map.yaml`



小车的位置

正常结果
出现小车和地图信息

如果报错命令找不到，或者命令出错，试试：
`sudo apt-get install ros-noetic-navigation`



导航功能的使用



2D Pose Estimate

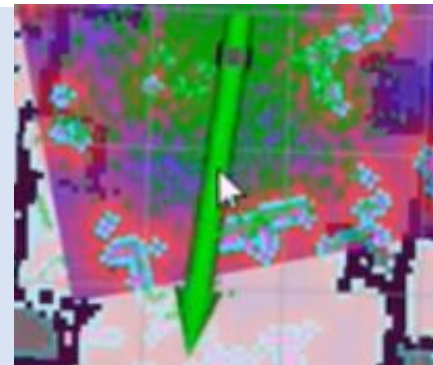


2D Nav Goal

操作：确定机器人相对于地图所处的位置和初始姿态：

点击工具栏**2D Pose Estimate**按键。

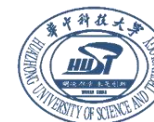
在地图上点击机器人相对于环境的正确位置并拖动箭头以确定机器人的姿态，可以通过多次调整确定最合适的初始位置设定。



操作：设定目标位置和目标姿态，设定方法如下：

点击工具栏**2D Nav Goal**按键。

点击一个确定的点并拖动箭头以设定目标位置和机器人最后停止的方向。



导航功能的结果和注意事项

 2D Pose Estimate 2D Nav Goal

结果：确定好初始位置和目的位置及姿态之后，起点和终点间生成路径，机器人自行计算运行路径并移动至目标点。

注意事项：



在开始导航前，应尽可能多次拖动箭头来**精确确定小车的位置**，否则可能出现小车无法启动的情况，因为当构建的地图面积不大时，小车的起始位置的相对误差就会变大。

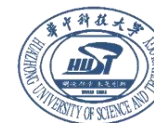
实验中**杂乱的周边环境**对导航功能会产生较大影响。所以总的来说，小车的运行原理和计算能力对实验场地提出了一定要求。

如果无法生成路线，考虑**设置更近的终点/调整起点位置**。



The background features a large, light blue downward-pointing triangle. Inside it is a smaller, solid blue downward-pointing triangle. A thin, light blue line forms an upward-pointing triangle that overlaps with the others. A thick blue arrow points from the upper left towards the center, and another thick blue arrow points downwards from the center. The text "THANK YOU" is centered in a bold, blue, sans-serif font.

THANK YOU



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